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Structured Credit Products

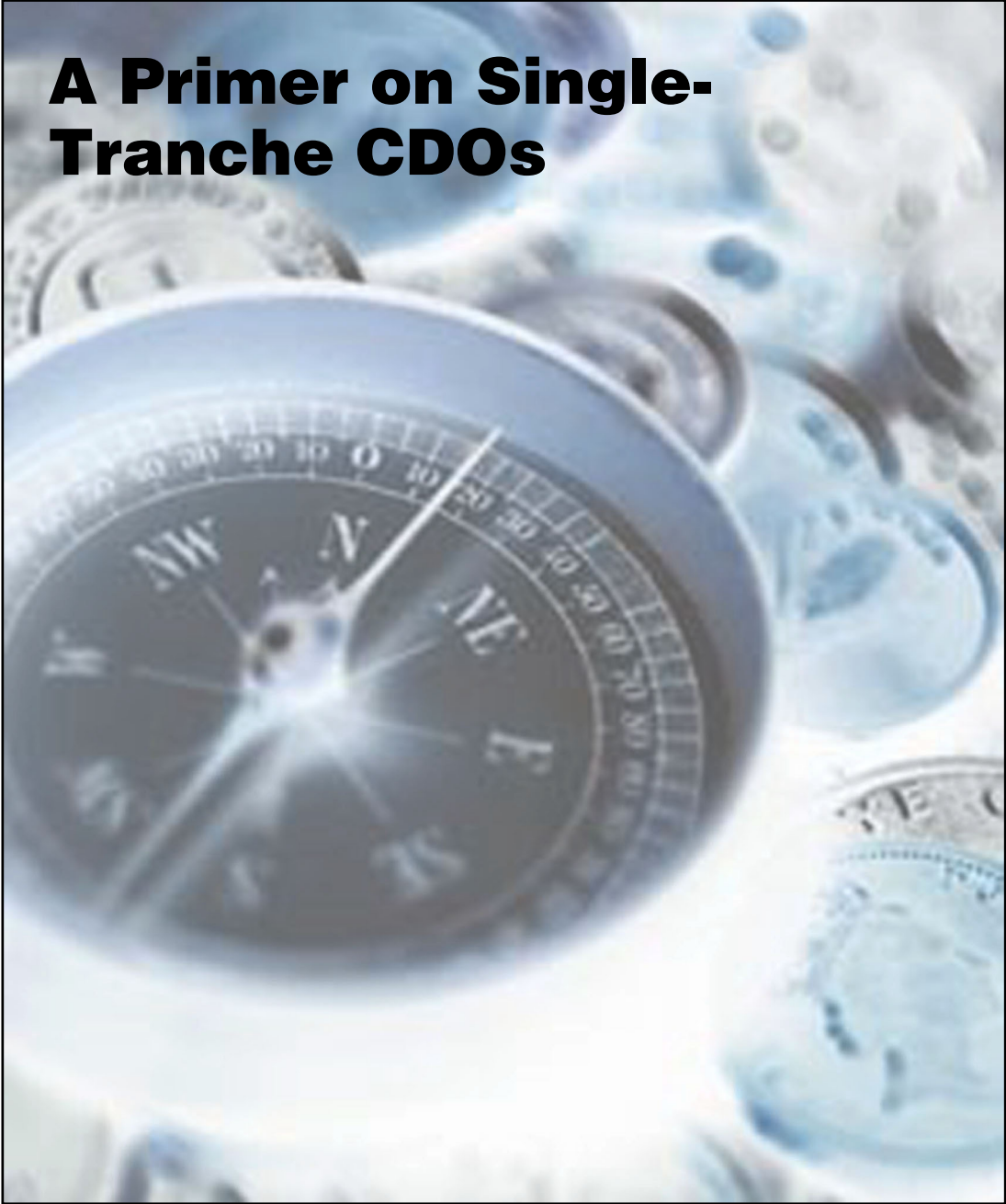
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A Primer on Single-Tranche CDOs



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A Primer on Single- Tranche CDOs

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Single-Tranche CDOs

The Single-Tranche CDO (STCDO) product represents the newest generation of tranching technology. It gives investors exposure to a customized slice (tranche) of the credit risk of a selected portfolio of reference credits. Although it is usually structured in an unfunded form as a portfolio default swap, it can also be designed as a funded obligation through a credit-linked note. The advent of STCDOs has improved the liquidity, transparency and flexibility of tranching products.

Decision Steps for Investors

Investors in STCDOs have the flexibility to choose the credit portfolio that they want exposure to, or they may reference the STCDO to the liquid iBoxx index. Investors' desired rating and return target determines their choice of subordination level and tranche size. In addition, investors can substitute underlying credits in the portfolio at the prevailing market prices throughout the life of a single-tranche CDO.

Key Issues in Modeling and Valuation

The two key elements in the modeling and valuation of structured credit portfolio products such as single-tranche CDOs are single-name default probabilities for the entities in the reference portfolio and the default correlation among the credits. The standard model that has been adopted by the market is the Gaussian Copula model, which, when combined with Monte Carlo simulation, generates correlated default time scenarios and a loss distribution for a STCDO tranche.

Risk Measures, Hedging, and Substitution of Credits

Investors primarily focus on five risk measures: credit spread sensitivity, default sensitivity, convexity, default correlation, and time-decay sensitivity. Junior tranches have greater credit spread sensitivity and default sensitivity than senior tranches, while junior and senior tranches have opposite convexity properties. An increase in default correlation is positive for junior tranches and negative for senior tranches. Time decay is caused by mismatches between the changing rate of default-related losses and the (constant) premium stream.

The ability to substitute credits allows investors to stave off imminent credit events, or to tweak the yield and risk characteristics during the life of the STCDO. While the exact cost of substitution is specified by the copula model, a simple approximation lets investors ballpark the fair value of substitution.

Single-Tranche CDO Investment Strategies

Single-tranche CDOs are popular among investors because they can be used to execute a variety of customized investment objectives and strategies. Single-tranche CDO strategies can be classified into several categories: leverage, market view, correlation, relative value, and micro/macro hedging. We describe each strategy, explain the market motivation for each, and indicate the typical investor who is likely to employ a given strategy.

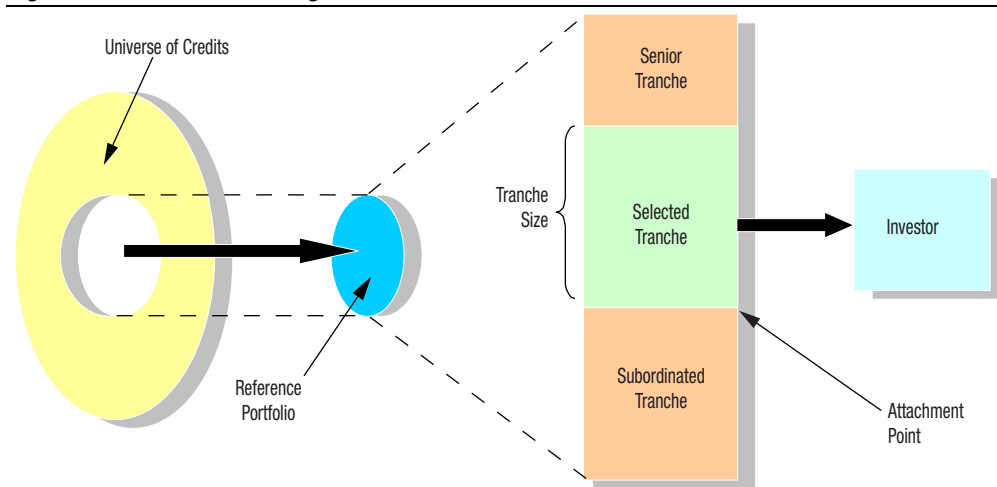
Single-Tranche CDOs

Overview

Single-tranche CDOs allow investors to take an exposure to a specific tranche of portfolio credit risk.

A single-tranche CDO (STCDO) is a powerful and flexible vehicle for investors seeking credit exposure consistent with their risk-return preference. The single-tranche CDO gives investors exposure to one slice (tranche) of credit risk of a selected portfolio of pre-selected reference credits. Investors may choose the credit portfolio that they want exposure to as well as the specific part or tranche of the capital structure. Investors can also choose a customized subordination level and tranche size. Investors with a higher rating requirement and a lower yield target can choose a greater level of subordination and/or a larger tranche size. The flexibility of the product is enhanced by investors' ability to substitute underlying credits in the portfolio at the prevailing market prices during the life of the investment (see Figure 1 for the basic structure of single-tranche CDO).

Figure 1. Basic Format of a Single-Tranche CDO



Source: Citigroup.

Advantages of Single-Tranche CDOs

Single-tranche CDOs provide multiple customized solutions for investors.

Single-tranche CDOs have several advantages versus alternative investment products. First, the leveraged tranche structure often provides higher yields than a similarly rated corporate investment. Second, investors have the ability to customize trades in terms of portfolio selection, the choice of subordination level, and tranche size. STCDOs also provide efficient market access to credits that might be expensive and/or difficult to acquire in the underlying cash markets. STCDOs give investors the ability to easily and efficiently go short the selected portfolio, sectors, and specific tranches. Investors can dynamically manage their positions through substitution of credits in the reference portfolio through the life of the single-tranche CDO. Finally, using single-tranche CDOs, investors can more efficiently hedge their portfolios, separating the fundamental (default) risk from the market-wide risk factors.

Key drivers for single-tranche CDOs are credit quality of individual entities in the portfolio, subordination level, tranche size, and the correlation structure.

Key Features of Single-Tranche CDO Transaction

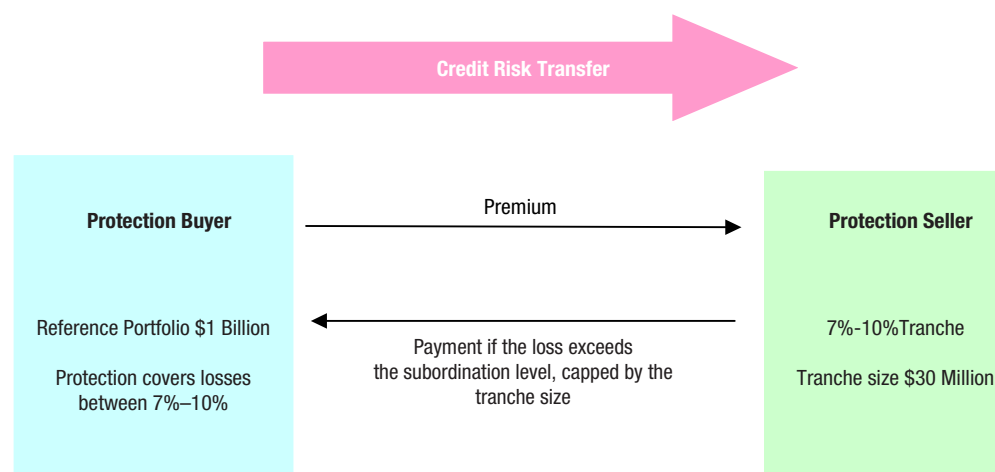
The key value drivers of single-tranche CDOs are the credit quality of the underlying portfolio, the subordination level and tranche size, and the correlation structure among the credits in the portfolio. Other features, such as substitution rights, also play an important role. Single-tranche CDO investors range from buy-and-hold investors who desire limited substitution rights and tend to hold the investment to maturity to correlation traders who use single-tranche CDOs to pursue various long and short strategies. However, traders are not the only ones who use STCDOs as defensive vehicles; for example, even traditional long-only investors can and do buy protection on single-tranche CDOs referenced to customized credit portfolios or the liquid iBoxx index to hedge fundamental (default) risk in their portfolios or to protect against market-wide spread selloffs.

Dealers take on risk when selling single-tranche CDO transactions to investors because they are placing only a portion of the capital structure. Dealers mitigate this risk by hedging with single name default swaps, default indices (e.g., iBoxx) or with other portfolio transactions. For example, “delta hedging,” offers dealers a way to manage their mark-to-market risk from underlying credit default swap spread movement. There are other risks that dealers need to manage as well, such as the risk of individual credits defaulting, correlation risk, shortening of maturity and convexity effects.

A single-tranche CDO is a contract between protection buyer and protection seller.

Description of the Product and Basic Structure

A single-tranche CDO is a financial contract between two parties that references a portfolio of credits. The parties are called protection buyer and protection seller. Once a reference portfolio is specified, a single tranche is commonly referred to as an “x excess y” tranche, where x is the tranche size and y the subordination level (attachment point). Equivalently, the tranche covers losses between y and x + y part of the total portfolio losses. Protection sellers can invest in a STCDO either in a funded note or unfunded portfolio default swap format. For ease of exposition, for the remainder of this report we will generally use the term *investor* to denote the seller of protection, and we will explicitly point out the instances where the investor is the buyer of protection.

Figure 2. Mechanics of Single-Tranche CDO Transaction

Source: Citigroup.

Single-tranche CDOs are commonly structured as unfunded portfolio swap . . .

Single-Tranche CDO — Unfunded Form

If structured as a portfolio default swap, the investor receives a periodic premium usually expressed as a fixed percentage in basis points of the outstanding notional amount of the tranche. In return the investor provides protection for any loss more than subordination level, but the loss payment made by investors is capped by the tranche size, i.e., the maximum loss for an investor is the tranche size.¹ Therefore, the cash flow exchanged between the two counterparties is default swap premium from protection buyer to the protection seller, and the loss payment, if any, from the protection seller to the protection buyer.

. . . and the same single-tranche CDO can also be structured as a funded obligation.

Single-Tranche CDO — Funded Form

Although the unfunded form is more typical, the same investment position can also be designed as a funded obligation through a credit-linked note. If structured in a funded note form, the notional amount the investor pays on closing is usually invested in high-quality, liquid assets such as triple-A rated asset-backed securities. The note pays fixed rate or LIBOR plus a premium on the outstanding notional. At maturity, the investor is paid back the notional, unless the losses exceed the subordination level. If that occurs, the notional is reduced, and a portion of the collateral is liquidated and paid to the protection buyer.

Variations of the Standard Single-Tranche CDO Structure

There are several variations to the standard single-tranche CDO transaction. First, variations can be created on how the premium is paid. For instance, the protection seller may prefer to receive the present value of the premium upfront or part of the premium upfront (this is referred to as a “points upfront” payment and is usually used in the equity tranches). Other protection sellers may prefer premium payments to be increasing or decreasing over time to mitigate or reverse time decay effects (see

¹ In a standard STCDO structure, it is easy to estimate the number of defaults in the underlying portfolio that will cause loss of payment by protection seller. As an example, assume that the tranche has 3% subordination, the reference portfolio has 100 names with \$10 million size each, and the recovery rate is 40% for each credit. Five defaults can happen without affecting the tranche (five defaults will lead to \$30 million loss on the portfolio ($5 * \$10 \text{ million} * (100\% - 40\%)$), and the losses associated with the sixth default will be covered by the protection seller.

the section “Single-Tranche CDO Risk Measures and Hedging” below). So far, most of the transactions have been based on a horizontal slice of the portfolio as represented in Figure 1, but there is no practical difficulty for dealers to offer a “vertical” slice of a tranche, such as 50% of the “3 excess 7” tranche, where the premium is paid based on *half* of the underlying notional amount, but the loss is based on the *full* amount of the tranche. Such tranche specification offers additional leverage and could be particularly interesting for investors selling protection on a pre-existing portfolio of credit obligations, where the original owner keeps the part of the “vertical” exposure in order to resolve potential adverse selection conflicts.

Second, the protection seller may want to have different parts of his investment exposed to different risk profiles. For example, the investor might want the final payoff of his investment principal linked to a single highly rated credit and the coupon payments to the junior STCDO tranche. Such variations, made possible by STCDO technology are similar in principle to a principal-protected structure, which allows risk-averse investors to enjoy the upside in a risky investment.

Third, the flexibility of single tranche CDOs allows investors to better manage their investment or exposures in conjunction with other business objectives. For example, a life insurance company might prefer to receive a decreasing premium payment scheme to increase its near-term earnings profile and better match its long-term asset-liability mix.

A single-tranche CDO can be structured in a fast and efficient way.

Ramp-up Period for Single-Tranche CDOs

The single-tranche CDO structure resolves another structural conflict inherent in traditional CDO investments. Traditional cash CDOs or balance-sheet synthetic CDOs are basically supply-driven products because brokers and dealers drive the reference portfolio selection and the ramp-up process. Unless they wish to carry a substantial amount of risk, the originators of a synthetic CDO are forced to sell the entire capital structure to investors. This process can be time-consuming and costly, often implicitly resulting in a conflict of interest between senior and junior investors. With only two parties to a single-tranche CDO, the buyer of a tranche (seller of protection) takes on the tranche credit risk and the protection buyer hedges mark-to-market risk in the liquid CDS market. This dramatically reduces the transaction period. Supply and demand for different parts of the capital structure are resolved through differences in the levels of implied correlation.

Main Decision Steps for Investors

In a typical single-tranche CDO transaction, there are three main decision steps for potential investors:

- 1 Select a portfolio of credits to that they want exposure to.
- 2 Choose a subordination level (attachment point) and a tranche size corresponding to their risk/return preference or yield target.
- 3 Dynamically manage their position and substitute credits in the collateral portfolio throughout the life of a single-tranche CDO.

Investors can select customized reference portfolios or use default index (e.g., iBoxx) to sell or buy default protection.

Selection of single-tranche CDO reference portfolio

The first step in structuring a single-tranche CDO transaction is the investors' selection of the credits in the underlying reference portfolio according to their preferences. For example, investors can choose a portfolio of credits different from their current positions, and by selling the protection on those names they achieve further diversification of their overall credit exposure. They can also sell protection on the subset of names in their current portfolio in case they want to overweight certain credit or sectors. Alternatively, investors can designate the whole or a part of their portfolios as the reference pool, and by buying protection they hedge themselves against spread widening or individual defaults.

As a common alternative, investors choose iBoxx default index, which is based on a diversified set of liquid names in the credit default market as a reference portfolio. The main reason for that choice is the liquidity and diversification of the index. The standard tranches on the iBoxx index also trade as liquid instruments in the broker/dealer market.

Subordination level and tranche size determine the leverage and protection premium.

Selection of the Subordination Level and Tranche Size

Once a portfolio is selected, investors must choose a subordination level and tranche size. The tranche size and subordination level determine the degree of leverage and the required protection premium. Investors that are primarily concerned about the rating and want to have the tranche rated by rating agencies such as Moody's and S&P, could choose the tranche size and subordination level so as to minimize the premium paid for the selected rating. Other investors could choose a subordination level that would provide a total spread equal to some desired return target. This "tranching" of credit portfolio risk can provide any desired risk/return profile. The idea is similar to one used by insurers for many years; the subordination level is equivalent to the deductible in the insurance contract and the tranche size is analogous to the concept of insurance risk ceiling ("maximum amount of coverage"). Also, by choosing the position of a tranche on a capital structure of STCDO, investors can separate their views on default risk from their views on market risk. We review such investment strategies in a later section. These trades are analyzed in more detail in our recent publication, *Bull and Bear in a Boxx*.²

In a single-tranche CDO transaction, investor have the ability to substitute the reference credits during the life of investment.

Substitution of Credits in the Reference Portfolio

The third distinctive feature of single-tranche CDOs is investors' ability to dynamically manage their investment gain and loss by substituting credits in the portfolio. Because the credit risk of individual credits in the portfolio changes after the initial transaction, investors might wish to substitute certain names. They could, for example, eliminate issuers that they perceive as potential credit blowups and substitute them with less risky names. Following each change, the dealer will adjust the premium for the tranche, change the level of subordination, or settle through a upfront payment to offset the impact on the substitution on the mark-to-market of the tranche. Substitution rights are especially important for market participants who intend to use single-tranche synthetic CDO structures as an efficient hedging and portfolio-rebalancing tool to manage their existing credit portfolio. In case the portfolio has experienced losses and

² *Bull and Bear in a Boxx — Using Tranching Products to Express Credit View*, Arvind Rajan, Graham Murphy, and Jure Skarabot, Citigroup, February 19, 2004.

the subordination below the invested tranches has been decreased, investors can improve the rating of their tranche by increasing the subordination level, even if they do not alter the composition of the reference portfolio.

Role of Outside Managers

Although substitution rights are inherently in the hands of the investors, some market participants prefer to delegate these tasks to an outside investors. Choosing a manager for a single-tranche CDO involves many of the same considerations that apply when picking a manager for a full capital structure synthetic CDO. In rating tranches managed by the third-party managers, the rating agencies pay special attention to the set of investment guidelines governing the managers' trading behavior. Investors should be careful to ensure that the guidelines and rules are set up to avoid a conflict of interest with the fiduciary responsibilities of the manager to the investor.

Key Issues in Modeling and Valuation

The cash flows of STCDOs depend only on the actual occurrence of defaults in the underlying credit portfolio. These cash flows are the overriding consideration for investors who are not subject to mark-to-market requirements and are expecting to hold the investment to maturity. However, investors who are subject to mark-to-market requirements, are also affected by the day-to-day spread movements of individual credit default swaps as well as the market perception of default correlation among the credits within the portfolio. Therefore, these two categories of investors would emphasize different risks when investing in STCDOs.

Two key issue elements for modeling and valuation are single-name default probabilities and default correlation structure.

Any STCDO pricing model has the following two essential parts:

- 1 A source of information that provides single-name default probability for the credits in the reference portfolio.
- 2 A mechanism for the introduction of default correlation among credits.

Single-Name Information

The most commonly used sources of information for single-name default analysis are:

- 1 Historical information based on rating agency default experience studies.
- 2 Market and rating-driven spreads for market-traded instruments such as credit default swaps.
- 3 Debt-equity based measures of default probability — for example, Moody's/KMV EDF model of PD (probability of default) from Citigroup's internal debt-equity model³.

The choice of information source should be driven by investment objectives.

Single-name default probabilities can be derived from various sources.

Historical information. This is used for performance and risk measurement by net risk takers interested in long-term credit exposure. However, these investors will tend to require additional premium over and above the compensation based on expected loss derived from historical data. This additional premium, much like “risk

³ See Jorge Sobehart and Sean Keenan, *Hybrid Probability of Default Models — A Practical Approach to Modeling Default Risk*, Citigroup, October 2003.

loading” within the insurance industry, is based on a multiple of unexpected loss. This risk premium is associated with the investor’s risk preference and appetite for unexpected losses.

Spreads. Some investors use a combination of rating and spread such as a matrix of idealized credit spreads classified by rating and industry. Others may use historical default data, but rather than using the current rating of the credit, they may choose to work with the rating implied from current market spreads.

Broker-dealers tend to use market information because they need to manage the composite trading book (portfolio transactions and single-name default swaps) in a consistent way.

Debt-equity information. Debt-equity based information of default probability incorporates forward-looking credit measures of default expectation for a specific firm over a given time horizon. The default expectation is based on the Merton-type structural model of default risk. This framework is driven by two main inputs: the asset volatility and the leverage of the firm. The asset volatility of the firm is determined from market observables, such as equity price and equity volatility, and leverage information based on the firm’s financial information. Commonly used debt-equity model measures are EDFs developed by Moody’s/KMV.

Default Correlation

Default correlation plays an important role in modeling and valuation of structured credit portfolio products.

The information on single-name default risk alone is insufficient to characterize the default risk of a portfolio tranche. This is because of default correlation. Put simply, default correlation measures the degree to which the default of one asset makes the default of another asset more or less likely. The concept of correlation is conceptually related to the more widely understood concept of diversification in credit portfolios.

Intuitively, one can think of default risk as being driven by a set of factors shared to different degrees by the individual credits. These factors tend to tie all credits into a common set of economic risks, but to different degrees. A particular factor may be associated with the general economy, or a region, or a specific sector, or, more idiosyncratically, related to the specific credit itself. Due to the shared influence of certain factors, it is generally believed that default correlation is positive between companies, even if they operate in different sectors. The greater the influence of general factors on a particular company, the greater the correlation of the default risk of that company with the general market, and with other companies in the market.

Within the same sector we would expect companies to have an even higher default correlation because they have more factors in common. As an example, consider the recent overcapacity of the telecommunication industry, a sector-specific issue, which resulted in the default of numerous telecommunication and telephone companies.

Quantifying default correlation is a challenge because of scarce data, even though the standard analytical framework is well understood. This framework, which is based on copula analysis, requires the prediction of joint (that is, correlated) survival times for all credits within a reference portfolio. Individual default probabilities only provide survival time for single credits. A copula function provides a method of obtaining a joint or multivariate distribution from a set of univariate distributions.

The Gaussian Copula model is the standard market approach to incorporate default correlation in structured credit portfolio modeling.

Model Implementation

Once a joint distribution of default times is obtained, a Monte Carlo simulation algorithm can be used to generate a large number of correlated default time scenarios. For each simulation scenario, we know exactly which credit defaults within the maturity of the transaction and exactly when that default occurs. By aggregating the results of all the scenarios, we estimate the loss distribution of the whole portfolio. The loss distribution of any particular tranche of a certain size with a subordination level can then be inferred from the portfolio's total loss distribution. The **break-even** spread for a tranche is defined as the premium that equates the present value of expected losses with the present value of premium payments for the tranche. This is the most common way of implementing the copula function approach to credit portfolio modeling. Alternative approaches that use Fast Fourier Transform, recursive calculation, or conditional analytical approximation can substantially reduce the computational time and noise inherent in the Monte Carlo simulation.

Role of Correlation

The expected loss of a portfolio of credits is the sum of the expected losses of each credit. This sum is not affected by portfolio diversity or default correlation.

However, while leaving its mean unchanged, default correlation does affect the shape of the portfolio loss distribution, particularly its variance and its tail. As tranches are exposed to a slice of the portfolio loss, they are particularly sensitive to properties of the corresponding range of the loss distribution. Thus, although the portfolio expected loss is indifferent, their expected loss is sensitive to correlation via its effect on the shape of the loss distribution.

Correlation estimates can be derived from historical analysis or can be used as market-implied factor for the quoted tranche prices.

Correlation estimates can be obtained from historical default analysis or they can be viewed as a parameter implied by quoted tranche prices. Intuitively, increasing correlation has the effect of making extreme events — very few defaults or very many defaults — more likely, thus increasing the loss variance and the loss tail.

The standard model that has been adopted by the market is the Gaussian Copula model.⁴ In this version of the model, correlation is incorporated via a parameter that is approximately the correlation of issuers' asset returns.

Risk measures (Greeks) estimate sensitivities of single-tranche CDOs to risk factors. Junior and senior tranches on capital structure can have different direction and magnitude of sensitivity.

Single-Tranche CDO Risk Measures and Hedging

Valuation models for single-tranche CDOs are important for more than just the pricing. They establish a platform to analyze the risks associated with the specific tranche investments. Based on the estimation of tranche risk sensitivities, buy-and-hold investors can estimate their mark-to-market exposure, and correlation traders can structure their hedging strategies. Note that different STCDO tranches on the capital structure can have a different direction and magnitude of sensitivity to each risk factor.

⁴ David Li: "On Default Correlation: A Copula Function Approach," *Journal of Fixed Income*, March 2000, pp. 43–53.

The buyer or seller of a single-tranche CDO should consider the following risks measures (Greeks)⁵:

- 1 Credit spread sensitivity (delta).
- 2 Credit spread convexity (gamma).
- 3 Correlation sensitivity (rho).
- 4 Default sensitivity (omega).
- 5 Time-decay sensitivity (theta).

Credit Spread Sensitivity — Delta

Delta measures the change in tranche mark-to-market value as the spread of one underlying credit moves.

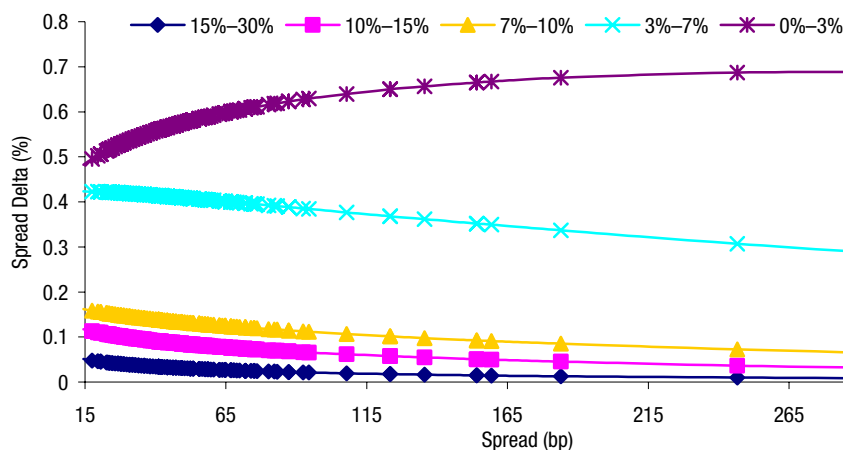
When the underlying credit spreads widen, the total portfolio expected loss will increase and correspondingly the expected loss of all tranches. A long position in tranches will thus see its mark-to-market decline and its breakeven spread rise as underlying spreads increase. There are two forms of spread sensitivity, individual (or micro) spread sensitivity *delta*, and sensitivity to a broad move in the portfolio spread, which is called the *Credit 01*.

We define the (individual) credit spread sensitivity *delta* as the ratio of the mark-to-market value change (ΔMTM) of a tranche T_j to the mark-to-market value change of a single name default swap when the spread of one individual credit CDS_i moves by a small amount, typically one basis point.

$$\Delta_i^{Tj} = \frac{\Delta MTM(Tj)}{\Delta MTM(CDS_i)}$$

Generally speaking, delta increases as we move down the capital structure, with the greatest delta (in absolute terms) in the equity tranche. In addition, spread delta for the equity tranche is higher for the individual credits with higher spread and spread delta for senior tranches is lower for the individual credits with higher spreads (see Figure 3). The reason is that when an individual credit spread is higher, the sensitivity of the senior tranche to this credit is lower as the specific credit is more likely to default and hit the equity tranche. For the same reason, spread delta for the equity tranche is higher for the credits with higher spread.

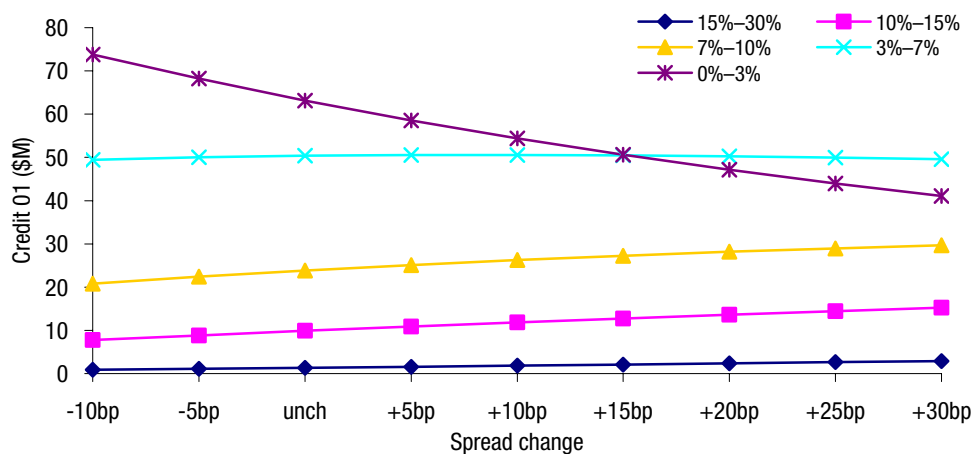
⁵ Default sensitivity has not yet been associated with a Greek letter. We use letter *omega* for default sensitivity.

Figure 3. Individual Credit Spread Deltas as a Function of Individual Credit Spread Levels

Source: Citigroup.

Tranche Credit 01 for parallel spread moves estimates the change in tranche value if all the names in the portfolio widen by 1bp.

In a similar fashion, we define the **Credit 01** as the mark-to-market change in dollar value of a tranche if *all the names* in the portfolio widen by 1bp in a parallel move. Credit 01 is an aggregate spread sensitivity measure, and is thus a suitable measure for estimating the hedge ratio when, for example, delta-hedging an iBoxx tranche with the underlying iBoxx index. Unlike an individual spread sensitivity, Credit 01 of senior tranches increases as all spreads widen in a parallel way and Credit 01 of the equity tranche decreases if all spreads widen in a parallel way (see Figure 4). With a parallel spread shift, the senior tranches become more risky and therefore more sensitive to spread widening, and the equity tranche becomes less sensitive to additional spread widening.

Figure 4. Credit 01 Versus Parallel Spread Changes

Source: Citigroup.

Credit Spread Convexity — Gamma

Just as with spread sensitivity, spread convexity comes in two flavors, macro- and micro (idiosyncratic or credit-specific).

Macro Spread Convexity

A long position in an equity tranche has positive spread convexity, and a long position in senior tranches has negative spread convexity.

We define macro spread convexity as the additional mark-to-market change on a tranche over that obtained by multiplying the Credit 01 by the parallel spread move for the underlying single-name credit default swaps. In general the delta to the portfolio spread is greatest in the equity tranche. However, if spreads widen on average, the expected loss on the equity tranche draws nearer to the maximum loss it can sustain — tranche notional. For this reason, a long credit position in the equity tranche will have positive convexity (like a long bond position), and a long credit position in more senior tranches will have negative convexity. The best way to understand this is through an analogy with how simple options work. The tranche loss profile with respect to total portfolio loss resembles the payoff profile at expiration of a call spread with respect to the price of the underlying. Buying protection on a senior tranche is thus equivalent to owning an out-of-the-money call spread on default-driven losses in the underlying portfolio. The first few defaults do not affect the payout of tranche. As spreads go up, defaults become more likely, so the out-of-the-money option comes closer to becoming an at-the-money option, and the tranche sensitivity to spreads increases. Conversely, protection bought on a sufficiently junior tranche is either at-the-money or in the money. Thus an increase in spread reduces its sensitivity, as it goes deeper in the money.

Idiosyncratic Spread Convexity

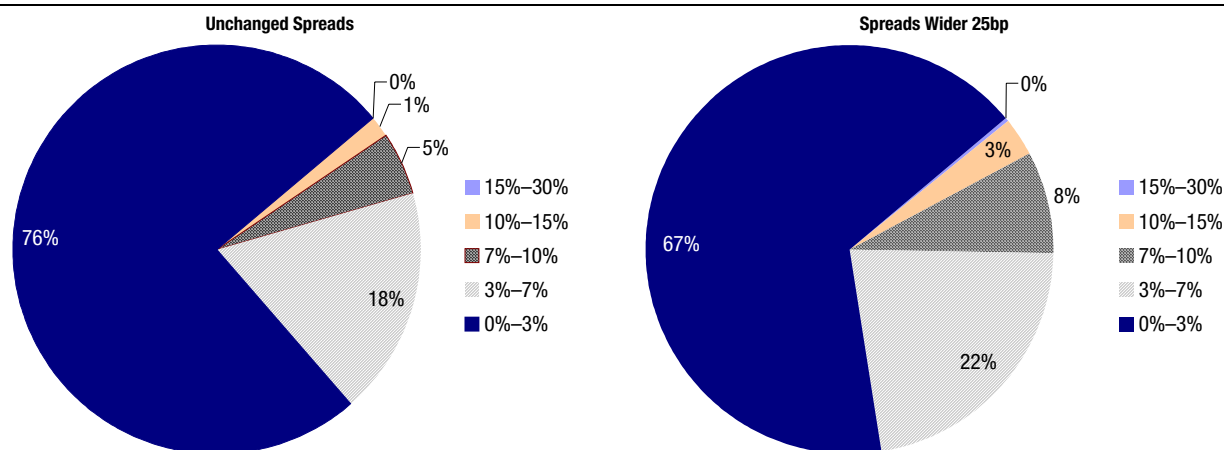
Perhaps against our intuition, the convexity of a senior tranche to the spread of an individual name in the portfolio is generally of the opposite sign to the convexity with respect to the average spread. But this makes sense if you consider how the sensitivity of the portfolio spread to the changes in the individual credit spread is distributed amongst tranches. We observe that wide credits have a greater impact on junior tranches, and tight credits have a relatively greater impact on senior tranches. As the spread of an individual credit widens, junior tranches receive a greater portion of the credit's delta and senior tranches receive less. Thus a long credit position in a senior tranche will have positive convexity to an individual credit spread.

Default Sensitivity — Omega

Equity tranches take the largest part of the default risk.

Following a credit event in the portfolio, the seller of protection on the equity tranche will compensate the buyer of protection for the loss suffered by the credit, and the notional amount of the tranche will be decreased by the one minus the recovery rate multiplied by the notional amount of the defaulted credit in the portfolio. The buyers of senior tranches will lose a portion of the subordination, and the tranches will become more risky. That will affect their mark-to-market positions.

We define **Default 01** as the mark-to-market change in the value of a tranche if one of the credits defaults (the defaulted credit is chosen to have the largest impact on the tranche). We observe that the equity tranche is allocated most of the Default 01 risk in the portfolio, but its percentage share decreases as the spreads widen (see Figure 5).

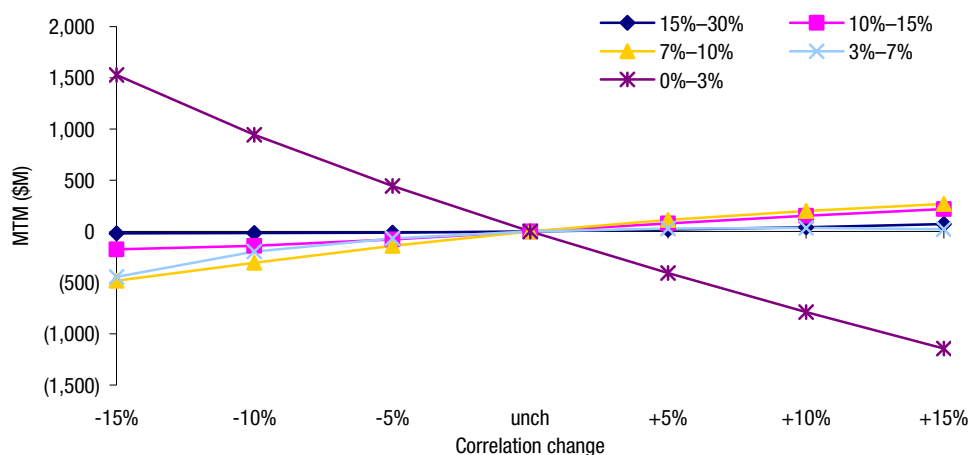
Figure 5. Distribution of Default 01 Risk Measure

Source: Citigroup.

Correlation sensitivity (rho)

Investors in senior tranches are short correlation, and investors in equity tranches are long correlations. Junior mezzanine tranche is usually correlation insensitive.

We understand that increased correlation makes more extreme events more likely. The increased likelihood of both “good” and “bad” extremes pushes expected loss out of junior tranches and into senior tranches, with the total expected loss unaffected. So, long equity tranche and short senior tranche credit positions have positive correlation sensitivity, and short equity and long senior tranche credit positions have negative correlation sensitivity. Also, with senior tranches gaining risk and junior tranches shedding it, there will also be a region in the capital structure that is relatively insensitive to a change in correlation. In a tight spread environment, this is generally the junior mezzanine tranche (see Figure 6).

Figure 6. Mark-to-Market Value Change Versus the Correlation Change

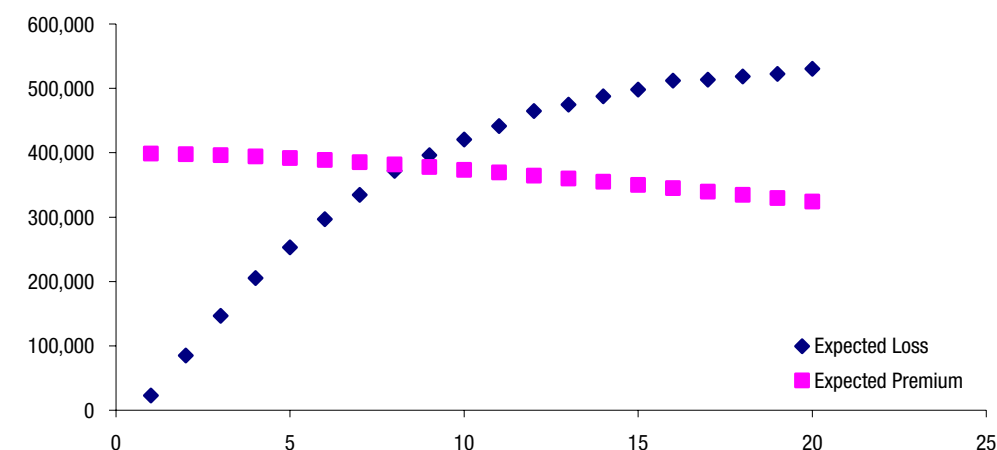
Source: Citigroup.

Theta measures the impact of time-decay on the value of the tranche. The value of the tranche changes as the time passes, because period by period, premium received is not exactly offsetting the expected tranche loss.

Time-Decay Sensitivity — Theta

Theta measures the impact on the value of tranches as the remaining time to maturity decreases. When we price a tranche, the breakeven spread is the spread, which would make the value of tranche loss equal to the total premium on expectation at inception. But as time elapses this equality will no longer hold. The reason is that the premium received for each period does not necessarily exactly offset the loss for the tranche in that period. The following graph shows the periodic expected loss and the expected premium over each period for the next five years for the iBoxx 3%–7% tranche (see Figure 7).

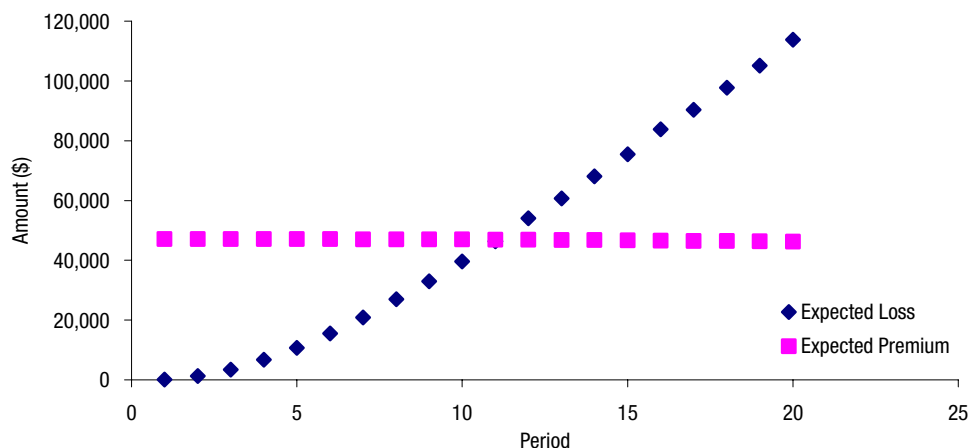
Figure 7. Periodic Expected Premium and Loss for Tranche (3%–7%)



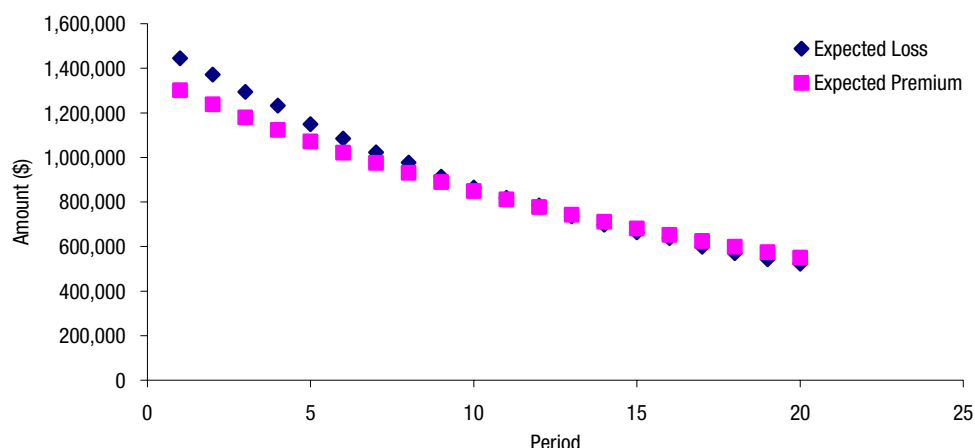
Source: Citigroup.

This graph shows that at the beginning the protection buyers pay more than enough to cover the expected loss over each period for the first nine periods. However, that relationship reverses after the first nine periods. So the theta for this tranche from protection provider's perspective is initially negative. This means that a long credit position in this tranche will initially suffer negative mark-to-market, like a premium bond. Similar graphs for other two tranches, 0–3% and 10%–15% are shown on the following figures (see Figures 8 and 9).

Figure 8. Periodic Expected Premium and Loss for Tranche (10%–15%)



Source: Citigroup.

Figure 9. Periodic Expected Premium and Loss for Tranche (0%–3%)

Source: Citigroup.

If paid as running spread, equity tranche premium is initially smaller than the expected loss. In the market, equity tranche default protection is usually paid as points upfront plus the fixed running premium.

For all senior tranches the periodic premium is flat, which reflects small incremental loss over each period while the expected periodic loss increases steadily. The equity tranche is the only tranche whose expected loss is larger than the expected premium at the origination of the transaction. In the above example (see Figure 9) the equity tranche premium is paid as the running spread. In most transacted cases, the market follows the “points upfront” convention, similar to the trading convention for the single-name CDS on the distressed names. In this case, the protection on the equity tranche is paid as percentage of the notional upfront and the fixed running premium (e.g., 500bp annual). Points upfront is defined as the difference between the present value of the expected loss on a tranche and the present value of expected fixed running premium. Therefore, as time passes, the value of points up front decreases, just as a bond priced initially at a discount pulls to par.

Substitution of Credits

Advantage of Substitution

Investors in single-tranche CDOs have the flexibility to replace credits in the underlying portfolio during the life of the tranche.

When investing in static CDOs, the investor’s knowledge on the underlying credit pool is only used at the time that the investor signs the contract. After that, the investor can only hope for the best performance of his investment with no option to dynamically manage credit risk by replacing credits of the underlying portfolio. In contrast, with the managed single tranche, the investor is allowed to make a limited number of credit replacements per year. For example, an investor might be allowed to substitute ten names per year, with a maximum change of 50% of the portfolio composition during the maturity period of five years for a single tranche transaction with a reference credit pool of 100 names. This flexibility gives the investor freedom to manage the investment in a dynamic way. For instance, managed single tranches avoided a portion of the losses associated with the recent default of Parmalat because certain investors made the substitution of Parmalat when Parmalat was traded with an expected recovery rate of 20 or 30%, while the ultimate recovery turned out to be far lower.

Settlement of credit substitution costs is done in a mark-to-market way as a difference in tranche values before and after the credit replacement.

Substitution Calculation

Since the dealer needs to re-hedge his position once a credit in the portfolio is replaced by another credit, the settlement of substitution has to be done in a mark-to-market way. The simplest way is to price the specific tranche with the replaced credit first, and then with the replacing credit, the difference being the substitution cost or benefit. This approach requires both parties of the transaction to agree on the model used and also the parameters used in the model. This has become possible since the Normal copula model has been accepted as the market standard for CDO pricing. The two parties usually agree on the pricing parameters, with a typical reference point being the current implied correlation embedded in the corresponding STCDO tranche. The cost of the substitution may be positive or negative — that is, paid by the tranche buyer to the dealer or from the dealer to the tranche buyer. The cost can be paid through any of the following:

- 1 Upfront payment between the protection buyer and the protection seller.
- 2 The alteration of the subordination level below the single-tranche CDO.
- 3 Resetting the coupon payable to the single-tranche CDO up to the point where the coupon has been reduced to zero, followed by either step (1) or step (2).

Each of the above methods of substitution settlement aims to offset the change in the mark-to-market value of the tranche before and after the substitution of the credit. The net effect on mark-to-market value of the credit substitution on one side and the cash payment, alternation of the tranche subordination, or change of the coupon on the other side should be zero. This mark-to-market change in tranche value can be captured by the net marginal credit factor.

The Marginal Credit Factors

The marginal credit factor can be used as an estimate of substitution costs.

The marginal credit factor (MCF) for each name and for each tranche measures the impact on the tranche value when the credit changes from risky status to risk-free status. It is in contrast to “jump-to-default,” and may be called the “jump-to-paradise” risk measure. This “jump-to-paradise” risk measure allows us to compare two credits in a portfolio context. The difference of MCF between the replacing credit and replaced credit is the substitution cost. That difference — net MCF — is equal to the change in the mark-to-market value of the tranche as valued with the old and new reference portfolio of credits.

MCFs tend to decrease when trading into tighter names because the value of protection provided by tranche decreases as the spread tightens. In this case, substitution results in the protection seller owing the protection buyer. MCFs also depend on the correlation of the name with the rest of the credit portfolio. Trading into lower correlated names increases the value of the protection provided by the equity tranche (and some lower mezzanine tranches), but decreases the value of protection by the senior tranche.

Approximate Substitution Cost Calculation

The problem with MCFs is that there is a need to have an agreement between dealer and investor on model, model parameters and also other model inputs, such as those single-name default swap spreads that are not easily verifiable. This lack of transparency leaves tranche investors dependent on dealer’s interpretation of risk and model parameterization. A simple way to ballpark the effect of substitution is to

consider the net impact of the substitution on the expected loss of the portfolio. We define the impact of substitution as the difference between the expected loss of the replaced credit and the replacing credit. Based on that calculation, we adjust the subordination level of each tranche to offset the net effect of substitution on the portfolio expected loss. Difference in expected loss of two individual credits can be calculated using Bloomberg CDSW page. For example, if we would like to replace a credit A with spread of x basis points by a credit B with spread of y basis points, then we calculate the difference in the present value of expected loss of credit A and credit B as a percentage of notional amount. After that, we reduce the subordination level of by that calculated percentage difference divided by the number of underlying credits in the portfolio. This approximation is based on the assumption that the net change of expected loss after substitution is deemed as if it is the actual loss incurred by the portfolio right away. Therefore, to make the adjustment for this net loss change, we lower the size of the equity tranche by this loss change, and make the same parallel adjustments in the subordination level of all other tranches on the capital structure.

This measurement ignores the difference in spread sensitivity of each tranche as well as the credit risk on the premium side. It usually underestimates the compensation required for the credit substitution. But the exact size and direction of the difference between the approximation approach and the mark-to-market approach depend on the various parameters, such as spread levels of the replaced credits, size and subordination of the tranche and the overall spread levels in the portfolio. But the advantage of this approximation approach is that investors can easily assess the substitution cost (in terms of substitution-level change) independently from a portfolio model output.

Trading and liquidity in single-tranche CDOs has risen dramatically because they have proven to be powerful and flexible structured credit products.

Single-Tranche CDO Market

Trading and liquidity in single-tranche CDOs has risen dramatically in recent months, since they have proven to be powerful and flexible structured credit products. The reason for their popularity is that investors can use them to execute a variety of customized investment objectives and solutions. Although these trading strategies are generally structured around the liquid market in iBoxx tranches, investors can apply the same strategies to customized portfolios in which they select the reference obligations of the single-tranche CDOs. As a combined approach, investors can start with standard iBoxx portfolio and substitute certain number of handpicked names for which they have preference not to have them included in their STCDO portfolio. As a representative example of single-tranche CDO, we present the liquid September 09 iBoxx tranches (see Figure 10). Each tranche trades at different implied correlation, which is market-driven factor in a standard copula model reflecting the demand and supply for the tranche protection. Implied correlation skew may also be due to copula model specifications and assumptions. We will explore correlation skew and its implications more fully in future research.

Figure 10. iBoxx Tranches, Each Tranche Notional Size is \$10 Million, 12 Apr 04

Tranche	Maturity	Implied Corr. (%)	Mid Spread (bp)	Credit 01 (\$000)	Default 01 (\$000)	Correlation 01 (\$000)
15%–30%	Sep 09	28.7	10.25	2.9	4.9	5.5
10%–15%	Sep 09	22.9	43.5	10.7	28.1	15.3
7%–10%	Sep 09	19.6	107.5	23.2	86.0	22.8
3%–7% ^b	Sep 09	—	295	45.5	282.9	0.7
0%–3% ^a	Sep 09	19.8	39%	65.4	1,395.9	-81.8
iBoxx ^b	Sep 09	—	54.5	4.9	47.5	0.0

^a 0%–3% iBoxx tranche is quoted in points of upfront payment plus a fixed 500bp running premium. ^b Tranche is correlation insensitive. Credit 01 is the mark-to-market change in dollar value of a tranche if each of the names in the portfolio widens by 1bp. Default 01 is the mark-to-market change in the value of a tranche if one of credits defaults (defaulted credit is chosen to have the largest impact on the tranche). Correlation 01 is the mark-to-market change in the value of the tranche if correlation changes for 1%. Risk figures are based on the implied correlation for each tranche.

Source: Citigroup.

Investment Strategies

Investors can use single-tranche CDOs to structure various investment strategies to target their risk/return profiles or hedging needs. We classify the strategies as:

- 1 Leverage strategies: Enhancing yield through leverage provided by tranching.
- 2 Market view strategies: Expressing a long or short view on the market using the appropriate tranches.
- 3 Correlation strategies: Expressing a view on implied correlation (equivalently, expressing a view on tranche technicals)
- 4 Relative value strategies: Taking a view on the relative cheapness of old versus new iBoxx tranches, five-year versus ten-year iBoxx tranche spreads, etc.
- 5 Micro/macro hedging strategies: Hedging individual default (micro) risk and/or market spread (macro) risk.

Leverage Strategies

Figure 11. Single-Tranche CDOs — Leverage Strategies

Strategy	Sample Trade	Motivation	Investors
Long position (sell protection) in mezzanine or senior tranche of iBoxx or customized portfolio.	Investor sells protection on \$10 million Sep 09 7%–10% iBoxx tranche. Tranche spread (mid): 107.5bp iBoxx spread (mid): 54.5 Date: 12 Apr 04	Investors who look for a buy-and-hold investment can obtain enhanced yield through leverage and express a bullish view.	Portfolio managers, CDO investors.

Source: Citigroup.

Long Position in Mezzanine or Senior Tranche

Investors can take a long buy-and-hold position (sell protection) in mezzanine or senior tranches on iBoxx or in customized portfolios. This trade should appeal to investors who are looking for enhanced yield through leverage. The motivation for this bullish trade is to obtain a targeted degree of leverage on the underlying portfolio while picking up spread respect to the iBoxx index or comparable corporate investments. Such buy-and-hold positions are often executed in an unfunded way, although investors can buy the credit through a funded credit-linked note, collateralized by triple-A rated paper.

A long position in a mezzanine or senior tranche allows investors to obtain leverage and express a bullish view.

Market View Strategies

Single-tranches CDOs that are referenced on iBoxx index products allow investors to separate their views on credit fundamentals (defaults) from their views on market and liquidity risk (spreads) and allow investors to take directional investment decision positions on defaults and spreads separately. Single-tranche trades are structured to be superior in terms of carry, leverage, and convexity to outright long or short market positions on the iBoxx index or a portfolio of individual names. We analyzed and compared several of such trades in our recent publication, *Bull and Bear in a Boxx*⁶.

Figure 12. Single-Tranche CDOs – Market View Strategies

Strategy	Sample Trade	Motivation	Investors
Short position (buy protection) in senior 7%–10% iBoxx tranche.	Investor buys protection on \$10 million Sep 09 7%–10% iBoxx tranche. Tranche spread (mid): 107.5bp iBoxx spread (mid): 54.5bp Date: 12 Apr 04	Investors who want to protect against spread backup can execute this leveraged, low-carry, positively convex position.	Portfolio managers, bearish investors that worry about a reversal in the corporate spread rally.
Long position (sell protection) in equity 0%–3% iBoxx tranche.	Investor sells protection on \$10 million Sep 09 0%–3% iBoxx tranche. Tranche spread (mid): 39 points upfront + 500bp running iBoxx spread (mid): 54.5 Date: 12 Apr 04	Investors that are bullish about default rates and want to enter into a leveraged, positively convex, high-carry position.	Credit-savvy hedge funds.
Carry- or delta-neutral combination of short position (buy protection) in 7%–10% iBoxx tranche and long position (sell protection) in 0%–3% iBoxx tranche.	Delta-neutral trade: Investor buys protection on \$28.2 million Sep 09 7%–10% iBoxx tranche. Investor sells protection on \$10 million Sep 09 0%–3% iBoxx tranche. 7%–10% tranche spread (mid): 107.5 0%–3% tranche spread (mid): 39 points upfront + 500bps running iBoxx spread (mid): 54.5 Date: 12 Apr 04	Investors that are bearish on spreads and bullish on credit quality can structure carry-neutral or delta-neutral position and maintain positive convexity.	Hedge funds and investors expressing view on global macro risk.

Source: Citigroup.

Short Position in Senior Tranche

Investors that are bearish on overall market-spread movements driven by global factors can take a short position (buy protection) in the senior iBoxx tranche. For example, we recommend buying protection on the 7%–10% iBoxx tranche because this tranche requires a lower premium for protection per unit of spread risk than junior tranches or an outright short with the iBoxx index. This position offers approximately five times the leverage and positive convexity relative to a short on the iBoxx index, and it provides a low carry protection against the spread backup.

Using single-tranche CDOs, investors can separate their views on default and spread risk.

⁶ *Bull and Bear in a Boxx — Using Tranchet Products to Express Credit View*, Arvind Rajan, Graham Murphy, and Jure Skarabot, Citigroup, February 19, 2004.

Long Position in Equity Tranche

Investors that are bullish on the credit quality of a portfolio of names or about default rates risk in the market as a whole, can go long (sell protection) the 0%-3% tranche. This trade offers bullish investors a leveraged and positively convex position with a high carry. Of course, this trade is exposed to defaults and to spread widening. Depending on their fundamental view, investors can hedge certain individual credits in the reference portfolio by buying default protection on those names and reducing the idiosyncratic risk of the trade. After partially hedging against defaults, investors still should be left with a significant positive carry on the trade.

Carry- or Delta-Neutral Combinations of Short Senior Tranche and Long Equity Tranche

Investors that are bearish on the possibility spread selloff and bullish on credit quality can combine a long position in the equity tranche with a short position in the senior tranche to pay for the negative carry. This carry-neutral trade captures the effect of positive convexity for senior and junior tranche and generates gains in the case of larger spread moves. A short senior and long equity tranche position can also be combined into a delta-neutral trade. Compared with directional trades, this delta-neutral combination generates lower, but still positive gains in spread selloff and spread rally scenarios.

Market view strategies can be structured as carry or duration neutral trades, maintaining positive convexity.

Correlations Strategies

Using a single-tranche synthetic CDO, investors can express their view on the implied correlation of tranches as different tranches can have opposite reactions to changes in correlation. Investors can choose the tranche subordination and size in line with their view on correlation. Note that the implied correlation is primarily a market-based factor, driven by the demand and supply of protection for each individual tranche. In most cases, correlation traders delta hedge their positions against the spread risk. Name-specific spread risk (micro spread risk) can be delta hedged with the single-name credit default swaps, and market spread risk (macro spread risk) can be hedged by the appropriate position in the iBoxx index. In addition, correlation traders can also hedge a portion of the default risk with a set of selected single-name credit default swaps. Therefore, these trades allow correlation traders to take a view on market demand and supply for single-tranche CDOs, without taking on credit risk.

Correlation traders can take a view on market supply and demand for single-tranche CDOs.

Figure 13. Single-Tranche CDOs – Correlation Strategies

Strategy	Sample Trade	Motivation	Investors
Long Correlation Trade: Short position (buy protection) in senior tranche or long position (sell protection) in equity tranche. Delta hedge with single-name CDS or iBoxx index.	Investor buys protection on \$10 million Sep 09 7%–10% iBoxx tranche. Investor sells protection on \$47.3 million Sep 09 iBoxx IG index Tranche spread (mid): 107.5bp iBoxx spread (mid): 54.5bp Date: 12 Apr 04	Investors who want to have long correlation exposure and expect implied correlation to increase. Positive carry.	Correlation traders (hedge funds, bank proprietary desks, and broker/dealers) with a positive view on implied correlation.
Short Correlation Trade: Long position (sell protection) in senior tranche or short position (buy protection) in equity tranche. Delta hedge with single-name CDS or iBoxx index.	Investor sells protection on \$10 million Sep 09 7%–10% iBoxx tranche. Investor buys protection on \$47.3 million Sep 09 iBoxx IG index. Tranche spread (mid): 107.5bp iBoxx spread (mid): 54.5bp Date: 12 Apr 04	Investors who want to have short correlation exposure and expect implied correlation to decrease.	Correlation trades (hedge funds, bank proprietary desks, and broker/dealers) with a negative view on implied correlation.
Correlation-Insensitive Trade: Long (sell protection) or short (buy protection) position in junior mezzanine tranche.	Investor sells or buys protection on \$10 million Sep 09 3%–7% iBoxx tranche. Tranche spread (mid): 295 iBoxx spread (mid): 54.5 Date: 12 Apr 04	Investors that are uncertain about correlation levels and don't want to take correlation risk, but want to invest in single-tranche CDOs to get leverage or take a directional market view position.	Investors who don't want to take exposure to changes in correlation.

Source: Citigroup.

Short Position in Senior Tranche or Long Position in Equity Tranche (Long Correlation)

Investors, who prefer to be long correlation, can set up such a position by going short the senior tranche (buying protection) or going long the equity tranche (selling protection). A rise in senior tranche implied correlation would imply that increased demand for protection is pushing senior tranche spreads wider and, all else equal, making the trade profitable. A rise in equity tranche implied correlation would imply that increased supply of protection is pushing equity tranche spreads tighter and all else equal, making the trade profitable.

Long Position in Senior Tranche or Short Position in Equity Tranche (Short Correlation)

Investors can choose a short correlation position by going long the senior tranche (selling protection) or going short the equity tranche (buying protection). A decrease in senior tranche implied correlation would imply that the senior tranche is trading at lower spreads, making the trade profitable, with all else equal. A decrease in equity tranche implied correlation would imply that increased demand for protection is pushing equity tranche spreads wider, making the trade profitable.

Long or Short Position in Junior Mezzanine Tranche (Correlation Insensitive)

If investors are uncertain about default correlation levels, and they do not wish to take correlation risk, they can invest (sell protection) in the tranche on the capital structure that is correlation insensitive. Typically, this tranche would be the junior mezzanine (second loss tranche), but the exact attachment point and size depends on the characteristics of the collateral pool. Proactive investors can achieve a similar correlation-neutral position by investing in a more junior and a more senior tranche below and above the correlation inflection point. These two tranches have opposite sensitivities to the correlation, and investors can earn a higher spread with this position than investing in the correlation insensitive mezzanine tranche alone.

A junior mezzanine tranche is usually correlation insensitive, suitable for investors who do not want to take on the correlation risk.

Although at origination the specific tranche can be correlation insensitive, with time-decay the correlation sensitivity will change.

Relative Value Strategies

Figure 14. Single-Tranche CDOs – Relative Value Strategies

Strategy	Sample Trade	Motivation	Investors
Long/short position in on-the-run iBoxx tranches versus short/long position in matching off-the run iBoxx tranches.	Investor buys protection on \$10 million Sep 09 7%–10% iBoxx tranche. Investor sells protection on \$10 million Mar 09 7%–10% iBoxx tranche. Sep 09 7%–10% tranche spread (mid): 107.5bp Mar 09 7%–10% tranche spread (mid): 112.5bp iBoxx Sep 09 spread (mid): 54.5bp iBoxx Mar 09 spread (mid): 56bp Date: 12 Apr 04	Relative value trade opportunities at the same leverage level between the new Sep 09 iBoxx tranche and the old Mar 09 iBoxx tranche. Investors can focus on implied correlation as relative value measure.	Hedge funds, correlation traders, broker/dealers.
Long/short position in five-year iBoxx tranches versus long/short position in matching ten-year iBoxx tranches.	Investor sells protection on Sep 09 7%–10% iBoxx tranche. Investor buys protection on Sep 14 7%–10% iBoxx tranche.	Relative value trade opportunities between the five-year and ten-year iBoxx tranches. Investors use implied correlation skew surface as relative value measure.	Hedge funds, correlation traders, broker/dealers.
Long/short position in 3%–100% iBoxx tranches and short/long position in iBoxx IG index versus short/long position in 0%–3% iBoxx tranche.	Investor buys protection on \$10 million Sep 09 3%–100% iBoxx tranche. Investor delta-hedges by selling protection on \$6.8 million Sep 09 iBoxx IG index. Tranche spread (mid): 20 iBoxx spread (mid): 54.5 Date: 12 Apr 04	Relative value trade opportunities between the traded 0%–3% iBoxx tranche and synthetically replicated equity piece using 3%–100% iBoxx tranche and iBoxx index.	Relative value hedge funds.

Source: Citigroup.

Long/Short Position in On-The-Run iBoxx Tranches Versus Short/Long Position in Matching Off-The-Run iBoxx Tranches

The recent roll of the iBoxx indexes provided a set of relative value trade opportunities in the single-tranche market. With the two-sided market in on-the-run and off-the-run iBoxx tranches, investors can now compare matching tranches referenced to similar underlying portfolios. Investors can focus on the implied correlation of new and old iBoxx single-tranches and look for relative value opportunities at the same leverage level in the capital structure.

Long/Short Position in Five-Year iBoxx Tranches Versus Short/Long Position in Ten-Year iBoxx Tranches

Expecting that liquid market for ten-year iBoxx tranches will improve further, we believe that investors will be able to execute relative value trade strategies based on the comparison between the five-year and ten-year iBoxx tranches referenced to the same portfolio. Implied correlation skew surface should be an indicative cheap/rich measure for individual tranches.

On-the-run and off-the-run iBoxx tranches provide potential relative value opportunities.

Long/Short Position 3%–100% iBoxx Tranche With Short/Long Position in iBoxx Index Versus Short/Long Position in 0%–3% iBoxx Tranche

Investors can combine a 3%–100% iBoxx tranche and delta-hedge it with the iBoxx index to synthetically replicate the equity 0%-3% tranche of the iBoxx. Comparing the obtained spread premium with the market spread for the traded equity 0%-3% equity tranche of the iBoxx allows investors to enter into a relative value trade or execute the directional view by using a cheaper way to access the equity part on the iBoxx capital structure.

Micro/Macro Hedging Strategies

Figure 15. Single-Tranche CDOs – Micro/Macro Hedging Strategies

Strategy	Sample Trade	Motivation	Investors
Short position (buy protection) in equity tranche on a customized portfolio of names.	Investor buys protection on 0%–3% tranche of a customized portfolio. Investor sells protection to the same counterparty on individual CDS names (“exchange the deltas”).	Cheaper way to hedge against default risk than buying protection on each individual name.	Portfolio managers and bank loan portfolios.
Short position (buy protection) in equity tranche on a customized portfolio of names and long position (sell protection) in mezzanine iBoxx tranche.	Investor buys protection on 0%–3% tranche of a customized portfolio. Investor sells protection on Sep 09 7%–10% iBoxx tranche	Portfolio managers with a positive view on macro risk can compensate the cost of hedging by going long senior iBoxx tranche.	Portfolio managers or buy-and-hold investors that are not subject to mark-to-market accounting treatment.
Short position (buy protection) in senior iBoxx tranche.	Investor buys protection on \$10 million Sep 09 7%–10% iBoxx tranche. Tranche spread (mid): 107.5bp iBoxx spread (mid): 54.5bp Date: 12 Apr 04	Investors that already hedged the most risky names against default and want to hedge against market-wide spread widening. This strategy is more efficient hedge than a short iBoxx position.	Portfolio managers or buy-and-hold investors with strong fundamental approach to portfolio selection.
Long position (sell protection) in senior iBoxx tranche.	Investor sells protection on \$10 million Sep 09 7%–10% iBoxx tranche. Tranche spread (mid): 107.5bp iBoxx spread (mid): 54.5bp Date: 12 Apr 04	Portfolio managers that have bought default protection in single-name CDS market and want to protect their hedging portfolio against mark-to-market risk.	Bank loan portfolio managers.

Source: Citigroup.

Short Position in Equity Tranche

A short position in an equity tranche can be used as a hedge against defaults (micro hedging).

A short position (buying protection) in an equity tranche on a customized portfolio of names is an efficient way to hedge against default risk. Investors select names from their portfolio that, in their view, have high risk of default and the protection on the first-loss tranche of this portfolio. Usually, investors sell protection to the same counterparty on individual CDS names (“exchange the deltas”). Buying protection on the equity tranche of a customized portfolio is a cheaper way to hedge against default than buying protection on each individual name. In addition, investors that are comfortable with a certain number of defaults being unhedged can buy protection on the second-loss tranche. The higher attachment point of the tranche can substantially lower the cost of carry.

Short Position in Equity Tranche and Long Position in Mezzanine Tranche

Hedging against default using individual credit default swaps or a short position in the equity tranche can be an expensive strategy. Investors that are less sensitive to market spread widening or that have a positive view on the general macro risk factors but worry about defaults in their portfolio can compensate for their hedging cost by going long a senior tranche on the iBoxx. The premium received on the senior tranche protection should lower the cost of protection on the equity tranche.

Short Position in Senior Tranche

A short position in a senior tranche can be used as a hedge against market-wide spread moves (macro hedging).

Certain investors are searching for solutions that will protect them against general market spread widening in their portfolios, especially if they have already hedged the most risky names against default or they trust in their selection of credits based on their fundamental views. For such investors, hedging with the iBoxx is not the most efficient solution, because the outright market short does not separate default and spread risk protection. A more effective hedging strategy is to put on a short position (buy protection) in the senior tranche that has the highest Credit 01-to-carry ratio. We estimated that the 7%–10% iBoxx tranche provides investors with the most suitable hedge, after taking into account the liquidity component of traded iBoxx tranches.

Long Position in Mezzanine or Senior Tranche

A long position in mezzanine and senior iBoxx tranches is a leveraged position on market-wide spread movements. If investors, such as bank loan portfolio managers, have bought protection in the single-name CDS market against defaults in their portfolios, this hedging portfolio is exposed to mark-to-market risk driven primarily by macro factors. If investors want to protect against mark-to-market risk in their hedging portfolios, then a long position in the mezzanine or senior iBoxx tranche can provide the solution. Because the junior tranches take on relatively more default risk than senior tranches, they are much less suitable for such hedging purposes. In addition, a long position in the iBoxx index is a less efficient hedge against market-wide spread moves, because the index spread is affected as by the systematic market risk as by the credit-specific events.

Future Trends and Conclusion

Single-tranche CDOs are one of the important innovations among structured portfolio credit products. Extending their features and flexibility above the standard cash and synthetic CDO products, they allow investors to place various leveraged positions in the credit market and efficiently execute their investment strategies. Flexibility in choosing the reference portfolio allows investors to structure the portfolio investments that can be properly matched with their investment portfolios, as a separate investment strategy or as a hedging position.

It is our expectation that the single-tranche CDO products will remain an established component of the structured credit market. New products based on single-tranche methodology will become available, driven by the investors' focus on the alternative asset classes and the correlation between them. As one of the main trends we expect further standardization of single-tranche structures. As the default index linked tranches will trade under the same documentation, investors will be able to unwind their positions with any of the market makers. At the same pace, we expect that the derivative market on the standard single-tranches will develop to the mature stage.

Trading volume in the swaptions on single tranches referenced to the default index will provide the liquidity and opportunity to buy or sell volatility positions. We also expect that the forward market on tranching indices will expand, providing investors with further opportunities to express directional views.

We expect that the demand for single-tranche CDO will further increase with the growth of CDO of CDOs transactions. Single-tranche CDOs are flexible and suitable source of collateral. Currently, such transactions (CDO squared) already include single-tranche CDOs in their collateral pool. In search for yield, future CDO squared structures will source even larger pool of single-tranche CDOs.

The single-tranche methodology and structures are not unique to the liquid names in the investment-grade corporate universe. They can also be applied to other reference classes, from high-yield reference obligations to the whole variety of the structured finance products. Following the increased liquidity in high-yield and emerging markets default indexes (e.g., high-yield and emerging market iBoxx default indexes) and variety of other asset classes, we expect that in the future single-tranche CDOs will reference not only the corporate names, but a whole range of the market segments. These new product will provide investors with an efficient tool to structure their investments according to their views on macro trends and correlation between asset classes.

Disclosure Appendix

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